1. Control of Industrial Water Quality

Introduction

Water, once thought to be an inexhaustible resource, is becoming more and more scarce. Each year we hear of areas of the world where drought has caused water to be at a premium. When one considers that approximately seventy-five percent of the earth's surface is covered with water, it is difficult to conceive of the possibility that water is in short supply. However, the problem is not with the quantity of available water but rather with its quality. Only a small percentage of the total supply of water covering the earth is directly usable by man either for intake into a municipal system or for industrial purposes.

The fact that a supply of relatively pure water is important to industry is obvious considering the number of industries that locate along rivers and lakes. In fact, the existence of an adequate water supply is perhaps the most important criterion considered when determining the location of a new industrial plant. Aside from the use of water as a transportation medium, large quantities of water are used in industrial processes.

Among the many uses of water in industry are:

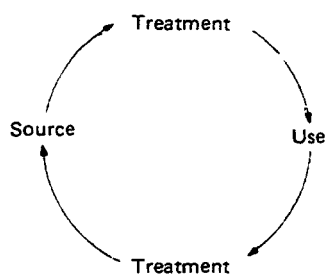
1. As a product additive.
2. As a source of heat and cooling of the plant (boilers and evaporative cooling units).
3. As a source of heat exchange to remove unnecessary heat from the industrial process.
4. In the sanitary services of the industrial plant.
5. As a major component in product cleaning.
6. As the most commonly used method for providing fire protection within the plant.
7. As an additive for the preparation of food and as a source of fluid for the workers.
8. In the general housekeeping and cleanup around the plant.

When water is used within the industrial plant, it can become polluted with waste products. These waste products include disease-producing organisms, inorganic and organic solids. Chemicals may be added to the water, either in a liquid or gaseous form. In some special cases, radioactive material can become suspended within the waste water from the plant. Finally, in cases where the water has been used as a source of heat exchange, the heat that has been added to the water causes its temperature to be unnaturally high and acts as a pollutant.

The typical water cycle that occurs in an industrial operation includes an original source of water, a treatment of water to meet the requirements of the industry, the use of the treated water in the process, and a post-treatment of the water in order that it may flow back to the source in a relatively harmless condition. The source of water that the plant may use can include lakes, rivers, streams, wells which have been drilled in the locality of the plant, or municipal water systems. The source that is used depends upon the availability of one of the above types of water supply and the quality of water that is needed. Often more than one source may be utilized in the plant as in the case where a municipal system supplies potable water while a local lake supplies water used in the plant processes and fire protection system.

Figure 8.1.1.

The water cycle.



The treatment of water prior to use in an industrial plant varies depending on the ultimate use of the water and the source from which it was obtained. The source may be treated, or its water may be usable in its original form. Certain additives may be required before the water can be used, such as in the case of boiler feed water where additives are used to prevent calcium buildup within the boiler itself. Any industrial operation requires a potable water supply for the workers to drink. Some processes require that potable water be used, particularly those where water is added to a product intended for human consumption such as in food processing.

In some processes and in fire protection, the need for potable water does not exist. However, some pre-treatment may still be necessary to remove certain chemicals and solids which are present in the raw water supply.

During process use, water becomes contaminated. It can no longer be used in the process and must be removed from the plant, or treated for reuse.

Prior to disposal, the waste water must be post-treated to remove any environmentally hazardous pollutants. In the past such treatment was not common, and the water was returned to the source untreated. For obvious reasons, this approach was not satisfactory in terms of the environmental state of natural water sources. In more recent times, environmental laws have

been enacted to control the pollutants that can be returned to the water source. Now, before the waste water can be returned to the source, it must be treated to remove potentially damaging pollutants. Such treatment may be accomplished by sending industrial waste water through a municipal sewerage system; or it may be necessary, depending upon the location of the plant and the type of pollutants in the water, that the water be treated in the plant before entry to either the source or a municipal sewerage system.

As water becomes more scarce, conservation of this precious resource is necessary. In some areas of the world, natural unpolluted water is at a premium today. Unless conservation methods are used, this situation will become more prevalent. The high cost of the treatment of source water can justify the need for conservation steps in the use of water in the plant. One method that is available to industry is the recycling of water within the plant. Using such an approach, water is recycled from one area to anotherunti1 the contaminants that have been added make it no longer possible to recycle the water without treatment.

Water Treatment

The treatment of water depends to a great extent on the use to which the water will be put. It also depends upon whether the treatment is of a raw water supply or of waste water. Similar unit processes are used for treatment of raw water sources and waste water. However, some additional procedures and care are necessary when water is being ed to provide a potable water supply. In addition, some specific tr ts may vary because of required additives or potential contaminants th present within the waste water.

Certain general unit process steps are required when treating water, either from a source or to remove waste materials. These process steps are:

1. Sedimentation to remove large solids from the water.
2. Control of the acidity or alkalinity of the water.
3. Destabilization and conglomeration; coagulation and flocculation to remove smaller suspended solids.
4. Filtration to further remove solids.
5. Bacterial digestion to remove organic materials.
6. Disinfection using chemical additives to control disease-carrying organisms.
7. Aeration to remove odors and gases, as well as to control organisms.
8. Disposal of waste sludge that has been removed from the treated water.

Each of these steps will be discussed in more detail in the following material.

Sedimentation. The first step that must be taken in the treatment of water either from a source or as waste water from processes is to remove the large solids in suspension in the water. This step is accomplished through the use of sedimentation.

In the sedimentation process, large solids settle out of the water if an adequate time period is available for such settling to occur. Often this

settling is accomplished by providing settling ponds, lakes, or tanks where the waste water or source water is pumped and held for a period of time.

As the water remains in these holding areas, the large solids tend to settle to the bottom of the storage area. Water is then removed from the surface of the tank for further treatment.

Settled sludge in the holding area must be removed occasionally or it will clog the holding tank or pond. In the case of a pond, periodic draining and removal of the sludge by bulldozers is necessary. Holding tanks often have sloping bottoms and are equipped with slowly rotating blades to push the sludge to the center where a small stream of water washes the sludge away.

In some cases, particularly in a municipal waste water plant, it is necessary to remove certain large solids such as waste paper, cans, etc., before further processing. In order to accomplish this, an initial screening removes the large solids, which, if allowed to continue through the system, might quickly clog the holding ponds or settling tanks.

Control of pH. The second step in the water treatment process is the control of the pH level of the water. This process step may be accomplished either before or after sedimentation. In general, pH control involves the treatment of highly acid water entering the treatment plant. This acid water must be neutralized in order to remove the potential for damage to piping and other equipment within the treatment plant. For this reason, pH control is often the first step in waste water treatment.

pH control for acid water is accomplished by adding alkali to the effluent. The most commonly used alkali additives are caustic soda, quicklime or sodium phosphate. These chemicals act to neutralize the acid in the water.

In many cases, pH control is required before waste water leaves an industrial plant to enter a municipal system. The municipal system will specify this control in order to protect the waste water collection piping system from potential damage.

Coagulation and Flocculation. The process of sedimentation removes large solids from the water being treated. However, many of the solids present within the water are so small that an extremely long period of time would be required for them to settle out. To remove these suspended solids, an approach is used to speed up the settling of the smaller solids.

The process of coagulation or destabilization which involves the addition of a chemical agent to cause the small solids to stick together to form larger solids, is used to accomplish this sedimentation. Small solids that have not precipitated during the original sedimentation process and which remain in suspension must be removed. The coagulant causes an electric or interionic force which makes the small particles of solids adhere when they come in contact with one another. Thus, larger solids are formed by the destabilization process.

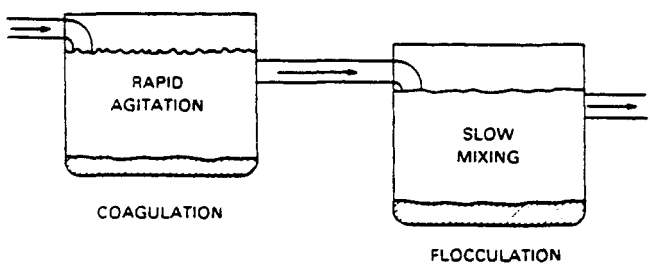
Coagulation is a fast chemical process, and as a result rapid agitation is desirable during this phase of treatment to increase the potential contact of

the small particles with one another. A common coagulant that can be added to the water being treated is alum, or aluminum salts.

Flocculation or conglomeration is the physical process in which the particles become enmeshed with one another. In general, this process is a slower process than coagulation. A gentle mixing or agitation over a longer period of time is necessary to cause the solids to group together or conglomerate. As the solids conglomerate, even larger particles are formed. As the particles become larger, the probability of settling is increased.

Figure 8.1.2

Coagulation-flocculation.



Generally, two tanks are used in the coagulation-flocculation process. The first tank, or the coagulant tank, introduces a high agitation to the water being treated as the destabilizer or coagulant is added. The water being treated remains in this tank for only a short period of time. Thesecond tank, the flocculation tank, involves a gentle mixing of the water over a longer period of time to allow for conglomeration and settling to occur.

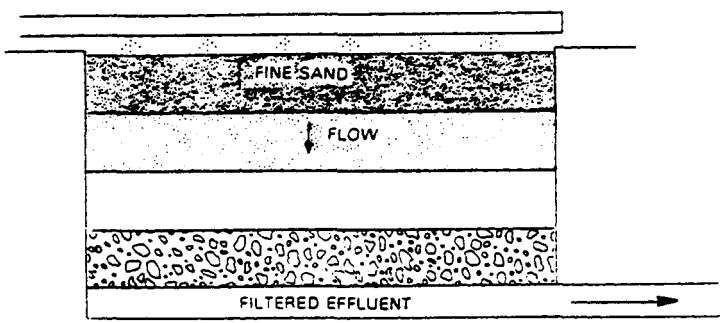
Filtration. After the conglomeration-flocculation process is completed, it is often necessary to treat the water further to remove suspended solids. In those cases where such further treatment is necessary, a filtration process is used. The filters used consist of sand and gravel beds. Coarse gravel is placed at the bottom of the filter, with finer mesh gravel and sand layers used to build up the filter. The finest sand is used on the top layer. As the water flows through this filter, the filter removes small suspended solids. As the filter is used, it begins to load up with solids and the flow of water through the filter slows. However, as the filter loads it becomes more efficient and more solids are removed. Finally, the flow of water through the filter slows to the point that the filter must be cleaned in order to maintain the ability of the plant to treat a sufficient amount of water.

In order to remove the solids built up with the filter, it is necessary to wash it. This washing is accomplished through the use of a backwash. Washing must take place before the filter becomes clogged or before a break through the fine layers can occur. Proper washing of a sand and gravel filter is

important to maintain the efficient flow and odor control function of the filtering process. Without proper washing, the probability of solids breaking through the upper layers of fine sand increases. When such a breakthrough occurs, the water being treated has direct access to the more coarse layers and fewer solids are removed.

Figure 8.1.3

Filtration.



Bacterial Digestion. Even after processing to remove large suspended solids, it is often necessary to include a step to remove organic solids that still may exist in the water. This is accomplished by bacterial digestion. In this process step, bacteria feed on the organic solids in the water, yielding carbon dioxide and proteins. The bacteria themselves need oxygen, a controlled temperature, and food (sludge) to live.

Two types of bacterial digestion systems are used. The first is an anaerobic process in which the oxygen supply is obtained from the materials in the sludge. Anaerobic digestion often occurs at the bottom of a settling pond. The second type is an aerobic process which uses oxygen supplied by the air. The aerobic process is accomplished by aerating the settling tanks and is faster than an anaerobic process. The aerobic process is often used when the organic load within the waste water is low.

Often, the waste water treatment process involves either an aerobic system or an anaerobic system followed by an aerobic system.

In conjunction with bacterial digestion, settling tanks are often used. These tanks utilize skimmers to remove floating solids and to clarify the waste water. The settling tanks provide for control of the system by removal of excess bacteria in order that the digestion system does not become overpopulated.

Control of Disease-Causing Organisms. If the water is to be used for a potable supply, it is necessary to control any disease-causing organisms that

may be present in the water supply. This is accomplished by the addition of chemicals that can kill the various organisms present. The most commonly used chemical in this case is chlorine. Chlorine is added to the water to provide both an immediate organism kill and a residual effect as the water is either stored for use in a potable water supply system or returned to the source.

Some other methods of disinfection have been tried and are used in particular cases. These include the use of ultraviolet light, heat, metal ions (silver), iodine, and ozone. In general, however, chlorine remains the most commonly used method for the control of disease-causing organisms.

Aeration. Aeration of the water being treated may be accomplished at various stages of the treatment. The purpose of aeration is to add oxygen and some other gases to the water. In addition, it can help to remove some of the gases that build-up during the treatment process, including carbon dioxide, hydrogen sulfide, and methane.

Various types of aerators are used. In some cases, a spray type aerator, in which the water is sprayed into the air in fine droplets, is used. Other methods include the waterfall and the multi-tray waterfall aerator. Finally, air may be introduced or bubbled into the treated water, as is often done during the bacterial digestion process.

If aeration is performed indoors, a potential hazard exists. The waste gases that are removed from the water during aeration (CO_2, H_2, and CH_4) can build up within a confined area. In order to remove these gases, it is necessary that adequate ventilation be provided to the enclosed area to prevent such a buildup from occurring.

Removal of Waste Sludge. The suspended solids removed from the water being treated result in a sludge that gathers at the bottom of the water treatment tanks of ponds. It is necessary to dispose of this sludge. Obviously this cannot be done by washing the sludge into the water source, since the treatment was performed to remove the sludge in the first place.

There are essentially two methods that can be used for removal of the waste sludge. The first method is incineration of the sludge. The second method is to remove the sludge to a landfill. In either case, it is necessary to remove excess water before the disposal of the sludge in order that a smaller volume is handled. Some potential does exist for reclamation of energy in a large industrial or municipal waste treatment plant by the incineration of the sludge. Incineration not only produces heat as well as various chemicals in the waste gases that may be potential energy, but also it has the advantage of reducing the volume of the sludge. Thus, a smaller quantity of sludge in the form of fly ash must be disposed of using the landfill process.

Other Water Treatment. In addition to the above general unit processes, it is necessary in some cases to provide special additives to water prior to its use. For example, in the case of general industrial process use of water it is often necessary to provide wetting agents and decalcifiers to the water before use in the process. In municipal water supply systems that treat water

to obtain a potable supply, fluoridation is often performed. Salt is often added to industrial water supplies, especially in areas of thermal stress.

Thermal Pollution

Often during an industrial process heat is added to the water. This heat becomes a pollutant and can cause environmental damage if hot waste water is returned to a natural source. Among the effects of thermal pollution are:

1. An increased odor of organic materials in the waste water.
2. Potential failure of subsequent waste water treatment, particularly that of bacterial digestion.
3. An increased growth of algae in the natural source.
4. Potential harm to marine life in the natural water source.
5. A lack of dissolved oxygen in the heated water.

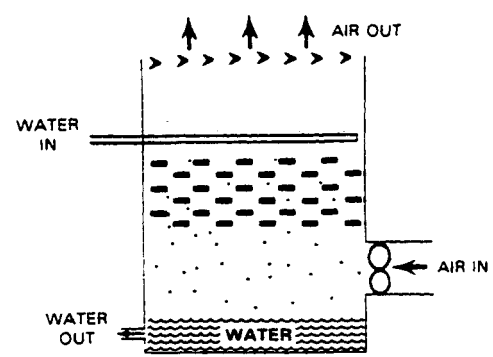
Two major methods of control can be used to remove the thermal pollution before entry to the natural source. These are the cooling tower and the spray pond.

The cooling tower is the most common method for removal of thermal pollution. In a cooling tower, water is pumped to the top of a tower and allowed to fall in a spray or a sheet of water through the tower. The tower contains baffles or decks that break the flow of water into smaller droplets. A source of upward motion of air through either natural draft or as a result of a fan drawing air through the tower at the top is available. The heat in the water is exchanged to the air through the evaporative cooling process.

The cooling tower is not without its problems. The evaporative cooling process produces a fog in the vicinity of the tower. If the water contains pollutants that are hazardous, these pollutants will be present in the fog. The effect of wind on the movement of the fog is termed "drift." This drift

Figure 8.1.4

Cooling tower.



can cause fog to move away from the cooling tower to the surrounding area. If the fog contains hazardous pollutants, these pollutants can affect the health of those living in the vicinity of the cooling tower. Thus, pollutants that are hazardous should be removed from the water prior to cooling in a cooling tower.

Cross Connection of Water Sources

Within an industrial plant, various sources of water may be used. A potable water supply is necessary for drinking as well as for use in products that will be consumed by humans. Water must be supplied for cooling of hot industrial processes. Water is also used for fire protection within the industrial plant. If water conservation is practiced, water will be recycled from process to process. Finally, when the water becomes so contaminated it can no longer be used, the waste water must be disposed of by removing this waste water to a treatment facility. Each of these various uses of water requires a distribution system between the source and areas of the plant.As a result, large quantities of water distribution piping are required. Unless precautions are taken, inadvertent cross connection of water sources can occur. This may result in the misuse of water containing hazardous contaminants.

One method for controlling water distribution to prevent the cross connection of sources is to label each of the distribution pipes as to the type of water that it carries. In addition, piping layouts that are properly labeled should be prepared and consulted when any connections are to be made.

In some cases it is desirable to have cross connection of sources. In a case where it is necessary to add makeup water to recycled water, a cross connection must be provided. Where it is necessary to connect the public water supply to a plant to provide for additional water which may be needed, a cross connection will be present. In these cases certain precautions must be taken in order to assure that the cross connection does not result in contamination of the source of cleaner water.

Contamination from such a cross connection can occur when a back pressure is developed that causes the contaminated water to flow back into the public water supply system. This back pressure can be as a result of many different causes. For example, a reduced pressure in one of the public water supply lines can result in contaminated water flowing from a higher pressure area back through the public water system. This reduced pressure can be a result of an interrupted supply, an excessive demand, a break in the public water supply lines, freezing, or improper sizing of the piping.

In order to control cross connections, it is necessary that the industrial hygiene engineer take certain precautionary steps. All piping should be reviewed for existing cross connections. Where possible, these cross connections should be eliminated.

A method for control is to use a movable connection when the supply of clean water must not be directly connected at all times. Thus, when required,

the movable connection can be attached to the source of potable water. Additionally, vacuum breakers and backflow preventers may be installed in the clean water supply line as an additional protection.

Control of Hazardous Materials in Water Treatment

The industrial hygiene engineer should be aware of the fact that some of the chemicals that are used in water treatment can be hazardous to the health of the workers exposed to them. For example, chlorine is well known as a hazardous chemical. The potential exposure to these hazardous chemicals of workers within the water treatment facility must be controlled.

The following table presents a selected list of chemicals used in the various water treatment processes. Special handling materials, threshold limit values, and flammable properties are indicated for these chemicals.

Table 8.1.1

Selected chemicals used in coagulation.

Name	Formula	Form	Handling	TLV	Flam.
Aluminum	$Al_2(SO_4)_3 \cdot 14H_2O$	Powder, lump, granular	dry-iron, steel; liquid-leadlined rubber, silicon asphalt, 316 stainless steel	Undetermined	—
Ammonium Aluminum Sulfate	$Al_2(SO_4)_3(NH_4)_2$ $SO_4 \cdot 24H_2O$	Powder, lump nut, pea	dureron, lead, rubber, silicon iron, stonewall	—	—
Bentonite	Colloidal Clay	Powder, pellet	iron, steel	Undetermined	—
Ferric Chloride	$FeCl_3$ $FeCl_3 \cdot 6H_2O$	Liquid, lump powder	glass, rubber, stoneware, synthetic resins	Undetermined	—
Ferric Sulfate	$Fe_2(SO_4)_3 \cdot 9H_2O$	Crystal, granule lump	ceramics, lead, plastic, rubber, 18-8 stainless steel	—	—

Table 8.1.1 (Continued)

Name	Formula	Form	Handling	TLV	Flam.
Potassium Aluminum Sulfate	$K_2SO_4AL(SO_4)_3 \bullet 24\ H_2O$	Powder, lump, granule	lead, leadlined rubber, stoneware	—	—
Anhydrous Ammonia	NH_3	Gas	glass, nickel, steel	25 ppm	yes
Chlorine	Cl_2	Liquified gas	glass, hard rubber, silver	1 ppm	yes
Ozone	O_3	Gas	aluminum, ceramics, iron, steel, wood	.1 ppm	—
Chloride Dioxide	ClO_2	Gas	plastic, soft rubber	.1 ppm	—
Sulfur Dioxide	SO_2	Gas	aluminum, brass, 316 stainless steel	5 ppm	—

Selected chemicals for pH control.

Name	Formula	Form	Handling	TLV	Flam.
Disodium Phosphate	$Na_2HPO_4 \bullet 12H_2O$	Crystal	cast iron, steel	—	—
Sodium Hydroxide	NaOH	Liquid, flake, lump	cast iron, rubber, steel	$.2mg/m^3$	—
Sulfuric Acid	H_2SO_4	Liquid	iron, steel	$1mg/m^3$	—
Trisodium Phosphate	$Na_3PO_4 \bullet 12H_2O$	Crystal	cast iron, steel	—	—

Summary

The treatment of water both prior to use in an industrial plant and after use to remove waste before entry into a natural water source is important in the industrial process. Without proper treatment of the water supply or waste water effluent, potential hazards are introduced to the work and community environments. The industrial hygiene engineer must be aware of the appropriate steps that should be taken to control this potential hazard. Assurance that proper treatment of waste water is occurring is an important part of the industrial hygiene engineer's job. In addition, the industrial hygiene engineer must be aware of any potential hazards that are introduced to the workers within the plant as a result of poor water quality or the contamination of potable water with a contaminated water supply through cross connection of sources.